

Agile Requirements— Can We Have Our Cake and Eat It Too?

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We must admit, neither of us really took to agile techniques. But nowadays it's agile this, agile that, agile everything. Many of our master's students use agile techniques in their work. Our information systems people are redesigning our website using agile techniques. Authors trumpet the benefits of agile requirements techniques. Can all these people be wrong?

So we decided to take another look at agile techniques and the requirements processes they support. This column reports what we found, and argues that we must rethink how to document requirements rather than avoid it.

A Short History of Agile

For all you youngsters out there, here's a brief history of agile. Over the last decade, software development has seen a substantial growth in the use of agile techniques. Agile emerged as an alternative way to develop software and manage projects. The rapid growth in its use has been impressive. Agile techniques are based on the 12 principles of the agile manifesto (www.agilemanifesto.org).

The manifesto states agile techniques should

- support people and their interactions, rather than use development processes and tools;
- deliver working software rather than documented requirements;
- enable direct collaboration with customers rather than negotiations about contracts that include requirements; and
- enable developers to respond to requirement changes rather than follow plans.

Unlike traditional methods that focus on modelling and analysis, agile encourages communication

and collaboration with end users to develop software without the need for modelling. Documentation that's large, paper-based, and difficult-to-read becomes the enemy because it impedes effective communication among people in projects.

So Is Agile Radical?

Looking back at the agile manifesto from current requirements practices, few of its elements are controversial. In fact, nowadays, it looks rather tame. For example, one principle invites us to welcome changing requirements, even late in development, because agile techniques harness change for the customer's competitive advantage. Few would dispute that requirements do change, or that ignoring such changes would lead to requirements that are out-of-date and don't describe what stakeholders actually want. Of course, analysts might differ on how to preempt requirements changes and manage them if they do occur. But analysts accept the need to handle requirements changes that the principle prescribes.

Writing Agile Requirements

Agile requirements techniques have probably made their biggest impact on how requirements are written and documented. Stakeholders write user stories, often in workshops. User stories are designed to be simple and small. Each describes a simple requirement, normally a user requirement, from a user's perspective. To keep each story short, they're designed to be written on a 3x5" card. Figure 1 shows several examples from a recent university project.

A project will often generate many user stories, all documented on different cards. This makes it easy to select and manage the requirements to implement. It also shifts the focus to communicating



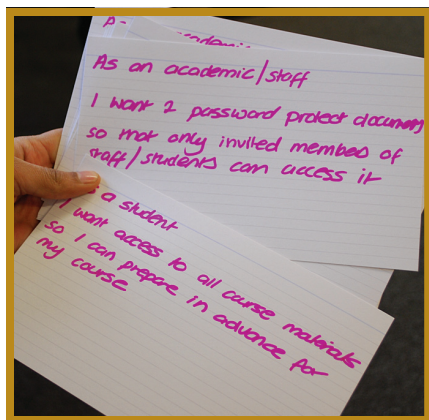


Figure 1. User stories. A project at City University London included examples of user stories—the most important requirements artifact in agile development techniques.

the requirements to be selected and implemented rather than more traditional requirements management activities such as version control and change management.

Agile Requirements Nirvana?

There's no doubt that agile techniques have been adopted successfully and have led to effective requirements practices. *IEEE Software* has reported on some of these stories and practices. For example, Lan Cao and Balasubramaniam Ramesh reviewed agile project requirements activities and reported that customers were able to discover requirements flexibly, steer their projects in unexpected directions, and use prototypes effectively to communicate during requirements processes.¹

But do these successes come at a cost? Probably. It seems clear that agile requirements techniques support some activities at the expense of others. For example, there's little support for discovering and analyzing business goals, describing and reasoning about nonfunctional requirements, or developing more innovative ideas because of short-term development sprints. Cao and Ramesh reported that projects neglected important nonfunctional requirements related to performance and security, selected inappropriate architectures, had to refactor software code, and didn't validate requirements with stakeholders before implementing them in sprints. Indeed, the lack of documentation made it difficult to evolve the application over time and induct new team members.¹



Figure 2. Digital conversation space. Developed in the JISC (Joint Information Systems Committee)-funded information spaces for collaborative creativity project, these spaces offer a vision for future requirements documentation.

These last few problems are critical. Agile development emerged as a response to document-led development, yet the absence of documentation has led to different problems. And let's be honest. These problems weren't difficult for the proponents of agile techniques to predict.

Rethinking Requirements Documentation

So, if requirements documentation was and still is a problem, why not rethink it rather than get rid of it? Why not produce documentation that supports, rather than impedes, requirements work? If documentation is too cumbersome and paper-based and impedes communication, then let's make it flexible, accessible and electronic to enable communication. After all, technology has come a long way since proponents posted the agile manifesto.

We've encountered similar challenges in our attempts to stimulate more creative thinking in requirements processes. One type of stimulus is the creativity workshop.² In a typical workshop, stakeholders use creativity techniques to discover, invent, and document as many concepts, requirements, and ideas in parallel as possible. The outcome of a workshop is substantial—hundreds of written ideas, sketches, visualizations, scenarios, and storyboards from which to generate requirements. We used to photograph and document workshop results in reports—large PDF files—which we'd make available on CD.

We shouldn't have been surprised when stakeholders and developers failed to make use of the workshops in subsequent requirements work.

So, we started to think about and ex-

periment with more dynamic and flexible electronic spaces for documenting requirements, along with new notations to communicate requirements. At the moment, we're exploring new technologies, including interactive tables, as well as low-tech solutions that make the use of technology as seamless and unobtrusive as possible. Figure 2 shows some recent examples developed in the JISC (Joint Information Systems Committee)-funded information spaces for collaborative creativity project.⁴ To support communication of and collaboration about requirements, we implemented *conversation spaces* that provide easy access to shared spaces that support familiar interactions such as sketching and writing with physical and digital artifacts. These spaces can be seen as digitally-augmented versions of the creative conversation spaces reported in the previous edition of this column.³ Early findings suggest that these spaces have the potential to support agile requirement activities without loss of documentation.

We originally wrote this column as a review of agile requirements techniques. It reminded us that agile was born, in part, out of too much poor documentation. While we recognize that documentation practices haven't changed dramatically over the last decade, we do believe that new technologies offer exciting new opportunities. Let's explore these new opportunities together, as a way to maximize the benefits of agile and other requirements techniques. 

References

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